In[ ]:= `CreateChapterTitleLA6Cloud[0, "Organizing and analyzing information"]`

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Introductory Example

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Quipu

Quipu

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The messengers of the Incas trekked through the Andes and into the coastal regions, carrying along sets of colored cords with knots known as quipu. The Incas controlled a vast region in South America and used quipu to keep track of the economic resources of the empire. The renowned magnificent cities and fortresses, predominantly built on highlands and on the steep slopes of the Andes Mountains, contained a sophisticated road system that bridged the various regions of their empire for the purpose of transporting goods and information. Roads paved in flat stone were the foundational fabric atop which products and ideas essential for human advancement were transported.

The versatility of the quipu recording system is remarkably impressive. Knots were tied with varying numbers of loops to represent the digits from zero to nine, and the spacing of the knots indicated whether a particular knot represented 1's, hundreds, thousands, and so on, making it possible to tie any number no matter how large. Think for a moment about the way different colored cords could be used to keep track of different commodities such as quinoa and alpacas. Using a system where strings can be tied onto other strings, one might consider using a main string for each village, with additional strings tied onto the main string to signify the different resources that belonged to that same village. Quipu were made of material readily available in a culture with well developed textiles and dyes, and they were easily transported from village to village.

Today the code of the quipu is somewhat of a mystery, leaving the complicated structures' meanings up to our imagination and puzzlement. Exactly what was being enumerated or what a particular color or organization of strings represented is unknown today. What we do know is that this method of counting and enumeration was well developed and there are conventions consistent across all surviving quipu, indicating that several people in the culture shared a common understanding of how this ingenious system worked. Without understanding the system today, though, we are unable to benefit from the wealth of information contained in the remaining quipu.

Numerous ways of counting and organizing data have been developed throughout history. Every culture has had some means of quantifying their surroundings, and the systems that were developed reflect the priorities and lifestyles of the culture. Mathematics is a uniquely human creation that has skyrocketed our ability to understand, enumerate, and control our world. Some groups counted on their fingers, resulting in a base 10 system such as that used by the Incas. Using fingers and toes resulted in a base 20 system, such as that used by the Mayans and some of the Inuit. Counting on knuckles might lead to a base 8 system. Creating larger numbers often involved addition and multiplication of smaller numbers or using zero a place holder. What is important for each of these counting systems is that the members of the communities all agreed upon the same conventions and understood how the system worked.

In mathematics, agreed upon conventions are referred to as *definitions or axioms*. Properties of the resulting system are then expressed as *lemmas* and *theorems*. The definitions that appear in textbooks have led to systems and theorems that are extremely valuable to present day people and are driving the technological revolution. In Chapter 0, matrices are introduced, which, like the quipu, can be used to record and preserve valuable information. There is a large amount of versatility in how to organize information. One very common method is to story it in a rectangular array. Related data can be recorded across a row or down a column. Decisions about how data is recorded will depend on what types of information are important to glean from it. .

In Chapter 2, we will begin to focus on ways to combine and process data. A convention for multiplying two matrices together is the first step. The chosen definition will seem cumbersome, at first, and even counterintuitive, but as we move on through Chapter 2 and the rest of the textbook, we will see that this particular way of multiplying matrices was chosen for its usefulness in processing, transforming, and extracting information. We will also develop properties of the mathematical system that results from our particular definition of multiplication and strategies that are useful when having computers calculate for us.

ORGANIZING DATA INTO TABLES

Organizing Data Into Tables

In as early as Grade 1, students are introduced to the idea of working with data in tables. In Example 1, a table listing the availablity of T-shirts at various store locations on the west coast of the United States is given. In this example, the numbers are kept small so that hand calculations will be straight forward. In this modern, data driven environment, understanding the processes applied to data is critical to understanding the output provided by various algorithms.

EXAMPLE 1

EXAMPLE 1

TABLE 1: T-shirt availability by color and location

Color/Location	Seattle	Portland	San Franciso	Los Angeles
Black	14	12	15	17
Grey	15	10	12	16
Blue	7	6	9	20
Crimson	12	8	10	12
White	7	8	3	6
Green	9	2	7	12

Horizontal strips of the data are referred to as the **rows** of the table and verticle strips are referred to as **columns**. Think about how information in the table might be combined. For example, adding all the numbers in a particular row calculates the number of T-shirts available of a particular color. In contrast, adding all the numbers in a particular column calculates the number of T-shirts at a particular location. More complicated questions can be asked about the data. For example, how much it would cost, for each color, to ship all the T-shirts to Seattle given that the cost of shipping from Portland to Seattle is $1 per T-shirt, from San Francisco to Seattle is $2 per T-shirt and the cost of shipping from Los Angelese to Seattle is $3 per T-shirt?

SOLUTION

This calculation can be thought of as multiplying each column by the cost of shipping from that location, and then adding up the corresponding information across each row:

Color/Location		Seattle		Portland		San Franciso		Los Angeles		Total Shipping Cost
Black		14		12		15		17		0 + 12 + 30 + 51 = 93
Grey		15		10		12		16		0 + 10 + 24 + 48 = 82
Blue	0	7	+ 1	6	+ 2	9	+ 3	20	=	0 + 6 + 18 + 60 = 84
Crimson		12		8		10		12		0 + 8 + 20 + 36 = 64
White		7		8		3		6		0 + 8 + 6 + 18 = 32
Green		9		2		7		12		0 + 4 + 14 + 36 = 52

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As the data, and the questions asked about it, become more sophisticated, the methods for combining and analysing the information become more sophisticated as well. Throughout this course, you will learn fountational material used to analyse and understand quantitative systems and various processes to analyse them.

Introduction

Notation

Throughout the text, it is important to pay attention to what different letters represent. If m and n are positive integers, an $\boldsymbol{m \times n}$ **matrix** is a rectangular array of numbers with m rows and n columns. If a matrix is labeled using the capital letter A, then the number or entry in the ith row and jth column of A is represented as a_{ij} and is called the (i, j)-entry of A. Similarly, the number or entry in the ith row and jth column of B is denoted by b_{ij}, and thus the individual entries of a matrix labeled with an uppercase letter can be easily be identified with the corresponding lower case letter subscripted with the row and column determining the position of the entry. The number of rows and columns of a matrix are listed as its **size**, with the numbers of rows listed first, followed by a ×, followed by the number of columns. The plural of matrix is **matrices**.

In the interactive figure below, us the setter bars to highlight different rows, columns, and entries of the 4×4 matrix illustrated in Figure A.

FIGURE A ◆: Denoting Matrix Entries

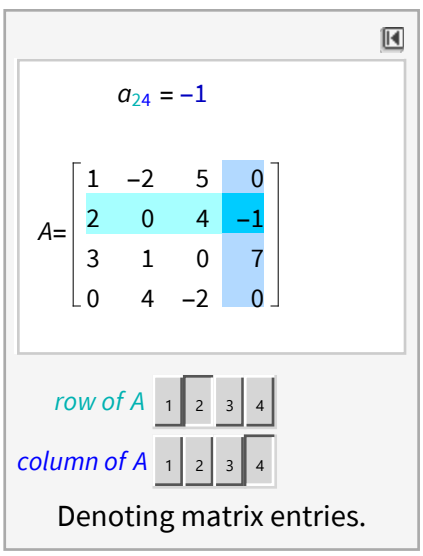

FIGURE A ◆

EXAMPLE 2

EXAMPLE 2

Let

$$A = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -9 & 3 \end{bmatrix} \text{and } M = \begin{bmatrix} 1 & 5 \\ -7 & -2 \\ 3 & 8 \end{bmatrix}$$

(a) Describe the sizes of the matrices A and M.

(b) Find the values of a_{23} and m_{31}.

(c) Describe the positions of the numbers -9 and 5 using letters with numerical subscripts.

SOLUTION

(a) Matrix A is 2×3, while matrix M is 3×2.

(b) Since a_{23} refers to the entry in row two and column three of the matrix A, $a_{23} = 3$. Similarly, m_{31} refers to the entry in row three and column one of the matrix M, so $m_{31} = 3$.

(c) Since -9 occurs in the second row and third column of A, $-9 = a_{22}$. Similarly, $5 = m_{12}$.

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It is often convenient to work with a *generic* matrix, one were the numbers are not actually known or represented. In this case, only letters with subscripts are used, but you can imagine the entries as numbers.

EXAMPLE 3

EXAMPLE 3

Write out a 4×2 matrix B using letters with subscripts as its entries.

SOLUTION

The matrix B needs to have four rows and two columns. Each entry can be described using a lower case b with the row and colunm numbers identified in the subscripts:

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \\ b_{41} & b_{42} \end{bmatrix}$$

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When the number of rows and columns of a matrix is not known, triple dots are used to indicate the missing entries in the description. For example, the $m \times n$ matrix A can be visualized as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Vectors in Rn

Vectors in $\mathbb{R}^n$

A **vector** in $\mathbb{R}^n$ is an $n \times 1$ matrix whose entries are chosen from the real numbers. Since vectors only have one column, the second subscript is not needed when denoting specific entries. Because vectors often play a special role in linear algebra, vectors in this

textbook will be denoted using a boldface non-italic font. For example, a vector $\mathbf{x}$ in $\mathbb{R}^2$ would be denoted as $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, whereas the vector $\mathbf{y}$ in $\mathbb{R}^4$ would be illustrated as $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$. In linear algebra, it is always important to pay attention to the context and what role each letter letter is playing in a description.

One of the important ways to view a matrix is to see it as a collection of vectors. For example, if $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$, then the n columns of A can be viewed n vectors in $\mathbb{R}^m$: $\mathbf{a}_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}$, ... , $\mathbf{a}_n = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$, in which case A can be also represented as $A = [\ \mathbf{a}_1\ \ \mathbf{a}_2\ \ \cdots\ \ \mathbf{a}_n\ ]$. Notice that in this instance the subscript on the $\mathbf{a}$'s represents the column number, rather than an entry in a matrix or vector.

EXAMPLE 4

EXAMPLE 4

Let

$$A = \begin{bmatrix} 1 & 5 & 3 \\ -7 & -2 & 2 \\ 3 & 8 & -2 \end{bmatrix}$$

(a) Write down the column $\mathbf{a}_2$ of A.

(b) For what value of n is $\mathbf{a}_2$ a vector in $\mathbb{R}^n$?

SOLUTION

(a) Just choose the entries in the second column of A and list them in a vector to get $\mathbf{a}_2 = \begin{bmatrix} 5 \\ -2 \\ 8 \end{bmatrix}$.

(b) Since $\mathbf{a}_2$ has three rows (or equivalently three entries), it is a vector in $\mathbb{R}^3$, so $n = 3$.

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EXAMPLE 5

EXAMPLE 5

Create a matrix B whose columns are the vectors $\mathbf{x}_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}$, and $\mathbf{x}_3 = \begin{bmatrix} 3 \\ -2 \\ -3 \end{bmatrix}$.

SOLUTION

Just list the vectors side by side to get the matrix $B = \begin{bmatrix} 1 & 2 & 3 \\ -2 & -2 & -2 \\ 3 & 2 & -3 \end{bmatrix}$.

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The Rows of a Matrix

The Rows of a Matrix

An additional way view a matrix and process the data it contains is to work with the rows of the matrix. For example, it may be helpful to visualize A as being organized into stripes of data:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

EXAMPLE 6

EXAMPLE 6

Let

$$A = \begin{bmatrix} 1 & 6 & 3 & -1 \\ -2 & -4 & 2 & 1 \\ 3 & 2 & -2 & 0 \end{bmatrix}$$

(a) Create a matrix B from A by multiplying all the entries in the second row by 2.

(b) Create a matrix C from A by interchanging rows 1 and 3.

SOLUTION

(a) $B = \begin{bmatrix} 1 & 6 & 3 & -1 \\ -4 & -8 & 4 & 2 \\ 3 & 2 & -2 & 0 \end{bmatrix}$

(b) $C = \begin{bmatrix} 3 & 2 & -2 & 0 \\ -2 & -4 & 2 & 1 \\ 1 & 6 & 3 & -1 \end{bmatrix}$

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Matrix Patterns

Matrix Patterns

Identifying patterns in matrices is an important component of analysing the systems or data they represent.

KEY WORD diagonal

The **diagonal entries** in an $m \times n$ matrix A are the entries where the two subscripts are match: a_{11}, a_{22}, a_{33}, …. They form the **main diagonal** of A. The main diagonal is highlighted for the $n \times n$ matrix A below.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

A **diagonal matrix** is a square matrix whose off-diagonal entries are zero. A particularly important diagonal matrix is called the **identity matrix,** an $n \times n$ matrix with ones down the diagonal and zeros everywhere else. The italicized letter I is reserved throughout the textbook to represent an identity matrix. When it is important to denote the size of the identity, since $m = n$, one subcript is used: I_n. For example, the 4×4 identity matrix is

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

An $m \times n$ matrix whose entries are all zero is a **zero matrix** and is written as 0. Again it is important to pay attention to the context. The symbol 0 may represent the number zero, an $m \times n$ matrix whose entries are zeros, or a vector (in bold non-italic font) all of whose entries are zeros. The size of a zero matrix needs to be determined from the context. A 3×5 zero matrix and the zero vector in $\mathbb{R}^4$ are illustrated below.

$$0 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \text{ and } \mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

An $m \times n$ matrix for which all the entries below the main diagonal are zero is called an **upper triangular matrix**. If all the entries above the main diagonal are zero the matrix is referred to as a **lower triangular matrix**. Consider

$$U = \begin{bmatrix} 1 & 0 & 3 & -4 & 1 \\ 0 & 0 & 2 & 3 & 1 \\ 0 & 0 & 4 & 5 & 4 \end{bmatrix} \text{ and } L = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 2 & 0 \\ 4 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Notice that U is a 3×5 upper triangular matrix and L is a 4×3 lower triagular matrix.

EXAMPLE 7

Is a diagonal matrix upper triangular or lower triangular?

SOLUTION

Since the entries below the main diagonal of a diagnonal matrix are zero, it is upper triangular. Since the entries above the main diagonal are also zero, it is a lower triangular matrix as well.

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PRACTICE PROBLEMS

PRACTICE PROBLEM

1. Write out an $n \times n$ lower triangluar matrix T, representing any nonzero entries with appropriate letters and subscripts, and listing zeros where appropriate.

2. Using the table from Example 1, create a 1×4 vector that describes the number of T-shirts available at each location that are either grey or crimson.

SOLUTIONS TO PRACTICE PROBLEMS

1. $T = \begin{bmatrix} t_{11} & 0 & 0 & \cdots & 0 \\ t_{21} & t_{22} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ t_{n-1\,1} & t_{n-1\,2} & \cdots & t_{n-1\,n-1} & 0 \\ t_{n1} & t_{n2} & \cdots & t_{n-1\,n} & t_{nn} \end{bmatrix}$

2. Notice that the number of grey T-shirts at each location are listed in row 2. The number of crimson T-shirts at each location are listed in row 4. Add them together to get [27 18 22 28], the number of T-shirts at each location that are either crimson or grey.

0.1 EXERCISES

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1. Let

$$A = \begin{bmatrix} 1 & 2 & 8 \\ 2 & 1 & 6 \\ -3 & 0 & 5 \end{bmatrix} \quad \text{and } B = \begin{bmatrix} 4 & 3 & 2 & 7 \\ 2 & -9 & 3 & -4 \end{bmatrix}$$

a. Describe the sizes of A and B.

b. Find the numbers in the positions a_{12}, a_{31}, b_{14}, and b_{23}.

c. Use the appropriate letters with subscripts to describe all the locations of the number 2.

2. Let

$$C = \begin{bmatrix} 2 & 3 & 8 & 3 \\ 4 & 1 & 4 & -1 \\ -3 & 0 & 5 & 7 \end{bmatrix} \quad \text{and } D = \begin{bmatrix} 4 & 6 & 2 \\ 2 & -9 & 3 \end{bmatrix}$$

a. Describe the sizes of C and D.

b. Find the numbers in the positions c_{14}, c_{32}, d_{13}, and d_{22}.

c. Use the appropriate letters with subscripts to describe all the locations of the number 3.

3. Represent a 3×2 matrix C using appropriate letters with subscripts as its entries.

4. Represent a 2×4 matrix A using appropriate letters with subscripts as its entries.

5. For the matrix A in Exercise 1, write out the vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ representing the columns of A.

6. For the matrix C in Exercise 2, write out the vectors $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$, and $\mathbf{c}_4$ representing the columns of C.

7. If the vectors $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 5 \\ 3 \\ -1 \end{bmatrix}$, and $\mathbf{a}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$ represent the columns of a matrix A, write out the matrix itself.

8. If the vectors $\mathbf{b}_1 = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$, $\mathbf{b}_3 = \begin{bmatrix} -5 \\ 3 \\ -1 \end{bmatrix}$ and $\mathbf{b}_4 = \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix}$ represent the columns of a matrix B, write out the matrix itself.

9. Write out a 3×3 diagonal matrix D, representing any nonzero entries with appropriate letters and subscripts, and listing zeros where appropriate.

10. Write out a 4×4 upper triangular matrix U, representing any nonzero entries with appropriate letters and subscripts, and listing zeros where appropriate.

For exercises 11 through 14, use the table in Example 1.

11. Create a 6×1 vector representing the total number of T-shirts of each color available in California.

12. Create a 6×1 vector representing the total number of T-shirts of each color available in Washington and Oregon combined.

13. Create a 4×1vector that describes the number of T-shirts available at each location that are either black or white.

14. Create a 4×1vector that describes the number of T-shirts available at each location that are either blue or green.